

Payal

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EDUCATION

Indian Institute of Technology, Roorkee
Bachelor of Technology - Chemical Engineering

Oct 2022 – May 2026
Roorkee, India

WORK EXPERIENCE

Data Engineer : Tredence Analytics

June 2026 – Present

- Built demand-pattern analysis on **100K+ daily pharmaceutical transactions** (PySpark, Databricks), projecting **20% savings in inventory holding costs**.
- Automated **PySpark** checks in **Bronze–Silver–Gold** ETL pipeline, flagging and quarantining **4.5%** of bad records.
- Reduced analytical query latency by **50.4%** by migrating normalized transactional data to a Star Schema and optimizing Delta tables for scalable analytical workloads.

Research Intern : Center for Sustainable Energy, IIT Roorkee

May 2025 – July 2025

- Built regression models on 5,000+ material records to predict the melting score of anti-icing pavement materials.
- Compared Random Forest and XGBoost using 5-fold CV; hyperparameter tuning reduced RMSE by 18%
- Ranked candidate materials by predicted melting score and used SHAP to explain the drivers

Software Intern : GlueX Protocol, Remote

Jan 2025 – March 2025

- Integrated 10+ DeFi APIs into a liquidity aggregator, expanding supported protocol coverage by **3x** overall.
- Resolved 8+ routing logic defects, improving **swap quote accuracy to 99.9%** across all integrations.
- Built **automated testing pipelines** with 99.9% coverage for stable and scalable protocol integrations.

PUBLICATIONS

Post-Audit Mirage: Identifiability Limits of Offline Agent Verification - Aman Behera, Payal Payal
NeurIPS 2026 Workshop VerifyAgents

Designed the matched-world benchmark spec and its formal evaluation contract (access layers, harm definitions, transport-gap estimand) for testing whether offline audits identify deployment safety.

PROJECT WORK

Personal Knowledge OS (April 2026):

- Architected agentic AI with autonomous tool routing; hybrid RAG(RRF + cross-encoder) for low latency
- Designed async pipeline: Celery + Redis queues + PostgreSQL checkpointing; scaled ingestion without blocking
- Deployed full stack: Chrome extension + FastAPI backend + Docker for reproducible production deployment

Roster Email Parser and Extractor (July 2025):

- Built a hybrid information-extraction pipeline combining OCR, rule-based parsing with NER for healthcare emails.
- Implemented validation, confidence scoring, and audit logging to ensure explainability and safe fallbacks.
- Improved performance via parallel processing increasing throughput by 50%, achieving 86% extraction accuracy.

Readthedocs (Sept 2025) [\[Live Demo\]](#):

- Built a end-to-end, Dockerized RAG pipeline over AWS docs, reducing hallucinations via chunk-grounded retrieval.
- Integrated HuggingFace embeddings with ChromaDB and FastAPI for scalable, source-grounded LLM inference APIs.
- Containerized the full stack using Docker, reducing setup time from hours to a single deployment command.

SKILLS

Generative AI (GenAI) & RAG: Retrieval-Augmented Generation (RAG), LangChain, FAISS, ChromaDB, Embeddings, Agentic AI Frameworks (LangGraph), NLP

Programming & Data Engineering: Python, Databricks/PySpark, REST APIs, MS SQL Server, ETL/ELT, Data Pipelines

ML Engineering: TensorFlow, PyTorch, scikit-learn, LangChain, LangGraph, XGBoost, Feature Engineering, Fine-tuning

MLOps & AI Observability & Cloud: Docker, Git, CI/CD pipelines, AWS (EC2), GCP, Linux